ARRAY ROTATION

"Implement array rotation (left/right) for arrays of size 10 by K positions without using extra space. Use a reversal algorithm. The program must validate K ($0 \le K < N$), show original → rotated → reversal steps visually, compute the sum difference, and support multiple rotations.

Sample Input/Output:

Enter 10 numbers: 1 2 3 4 5 6 7 8 9 10

Rotate Left (L)/Right (R): L

By positions: 3

Original: [1 2 3 4 5 6 7 8 9 10]

Step1: Reverse 0-2 → [3 2 1 4 5 6 7 8 9 10]

Step2: Reverse 3-9 → [3 2 1 10 9 8 7 6 5 4]

Step3: Reverse all → [4 5 6 7 8 9 10 1 2 3]

Sum difference: 0"

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>

void reverse(int a[], int start, int end)
{
    while (start < end)
    {
        int temp = a[start];
        a[start] = a[end];
        a[end] = temp;
        start++;
        end--;
    }
}

void printArray(int a[], int n)
{
    int i;
    printf("[");
    for(i = 0; i < n; i++)
    {
        printf("%d", a[i]);
```

OUTPUT

```
Enter 10 numbers:
Rotate Left(L)/Right(R): By positions: Original: [1 2 3 4 5 6 7 8 9 10]
Step1: Reverse 0-2 [3 2 1 4 5 6 7 8 9 10]
Step2: Reverse 3-9 [3 2 1 10 9 8 7 6 5 4]
Step3: Reverse all [4 5 6 7 8 9 10 1 2 3]
Sum diffference: 0
```

INPUT

1 2 3 4 5 6 7 8 9 10
L
3